

Q1. Attempt all questions

1. The portion of the operating system that always remains in memory is called the _____

✓ Answer:

Kernel

2. _____ scheduling selects the process that has been waiting for the longest in the ready queue.

✓ Answer:

FCFS (First Come First Serve)

3. In a multithreaded environment, each thread has its own _____, but shares code and data with other threads.

✓ Answer:

Stack

4. A _____ is a software that acts as an intermediary between a user and computer hardware.

✓ Answer:

Operating System

5. _____ is the phenomenon where a low-priority process holds a resource needed by a high-priority process.

✓ Answer:

Priority Inversion

6. The condition in which a process is waiting for a resource that is held by another waiting process is called _____.

✓ Answer:

Deadlock

7. In UNIX operating system, the command used to list files in a directory is _____.

✓ Answer:

ls

8. The method of allocating disk space where each file is a linked list of disk blocks is called _____ allocation.

✓ Answer:

Linked Allocation

9. In paging, the page number is mapped to a _____ number using the page table.

✓ Answer:

Frame

10. _____ is the process of removing a process from memory and storing it on disk when RAM is full.

✓ Answer:

Swapping

Q.2 Round Robin Scheduling (TQ = 2)

Given Table

Process Arrival Time (AT) Burst Time (BT)

P1	0	2
P2	1	3
P3	2	2
P4	3	2
P5	4	3

Step 1: Draw Gantt Chart

Time Quantum:

[

TQ = 2

]

Execution Order

- P1 executes from 0 → 2
- P2 executes from 2 → 4
- P3 executes from 4 → 6
- P4 executes from 6 → 8
- P5 executes from 8 → 10
- Remaining P2 executes from 10 → 11
- Remaining P5 executes from 11 → 12

Gantt Chart

0 2 4 6 8 10 11 12

| P1 | P2 | P3 | P4 | P5 | P2 | P5 |

Step 2: Completion Time (CT)

Process CT

P1	2
P2	11
P3	6
P4	8
P5	12

Step 3: Turnaround Time (TAT)

Formula:

$$[$$

$$TAT = CT - AT$$

$$]$$

Process	CT	AT	TAT
P1	2	0	2
P2	11	1	10
P3	6	2	4
P4	8	3	5
P5	12	4	8

Step 4: Waiting Time (WT)

Formula:

$$[$$

$$WT = TAT - BT$$

$$]$$

Process	TAT	BT	WT
P1	2	2	0
P2	10	3	7
P3	4	2	2
P4	5	2	3
P5	8	3	5

Step 5: Response Time (RT)

Formula:

$$[$$

$$RT = \text{First Start Time} - AT$$

$$]$$

Process	First Start Time	AT	RT
P1	0	0	0
P2	2	1	1
P3	4	2	2
P4	6	3	3
P5	8	4	4

Step 6: Average Turnaround Time

$$[$$

$$\frac{\text{Average TAT}}{\frac{2+10+4+5+8}{5}}$$

$$]$$

$$[$$

$$\frac{29}{5}$$

$$]$$

[
 $\boxed{5.8}$
]

Step 7: Average Waiting Time

[
 Average WT
 $\frac{0+7+2+3+5}{5}$
]
[
 $\frac{17}{5}$
]
[
 $\boxed{3.4}$
]

Final Answer

Process	CT	TAT	WT	RT
P1	2	2	0	0
P2	11	10	7	1
P3	6	4	2	2
P4	8	5	3	3
P5	12	8	5	4

Average TAT

[
 $\boxed{5.8}$
]

Average WT

[
 $\boxed{3.4}$
]

Q.3 Round Robin Scheduling (TQ = 3)

Given Table

Process Arrival Time (AT) Burst Time (BT)

P1	0	3
P2	1	3
P3	0	4
P4	3	3
P5	1	3

Step 1: Draw Gantt Chart

Time Quantum:

[
TQ = 3
]

Execution Sequence

- P1 executes from 0 → 3
- P3 executes from 3 → 6
- P2 executes from 6 → 9
- P5 executes from 9 → 12
- P4 executes from 12 → 15
- Remaining P3 executes from 15 → 16

Gantt Chart

0 3 6 9 12 15 16
| P1 | P3 | P2 | P5 | P4 | P3 |

Step 2: Completion Time (CT)

Process CT

P1	3
P2	9
P3	16
P4	15
P5	12

Step 3: Turnaround Time (TAT)

Formula:

[
 $TAT = CT - AT$
]

Process CT AT TAT

P1	3	0	3
P2	9	1	8
P3	16	0	16
P4	15	3	12
P5	12	1	11

Step 4: Waiting Time (WT)

Formula:

[
 $WT = TAT - BT$
]

Process TAT BT WT

P1	3	3	0
P2	8	3	5
P3	16	4	12

Process	TAT	BT	WT
P4	12	3	9
P5	11	3	8

Step 5: Response Time (RT)

Formula:

$$RT = \text{First Start Time} - AT$$

Process	First Start Time	AT	RT
P1	0	0	0
P2	6	1	5
P3	3	0	3
P4	12	3	9
P5	9	1	8

Step 6: Average Turnaround Time

$$\frac{3+8+16+12+11}{5}$$

$$\frac{50}{5}$$

$$10$$

Step 7: Average Waiting Time

$$\frac{0+5+12+9+8}{5}$$

$$\frac{34}{5}$$

$$6.8$$

Final Answer Table

Process	CT	TAT	WT	RT
P1	3	3	0	0
P2	9	8	5	5
P3	16	16	12	3

Process	CT	TAT	WT	RT
P4	15	12	9	9
P5	12	11	8	8

Average TAT

[
 $\boxed{10}$
]

Average WT

[
 $\boxed{6.8}$
]

Q.4 Find the miss and hit ratios using the Optimal Page Replacement Algorithm

Given

Frame Size

[
3
]

Reference String

0, 1, 2, 3, 3, 2, 1, 0, 4, 0, 5, 1, 6, 2, 3

Optimal Page Replacement Algorithm

In Optimal Page Replacement:

- Replace the page that will not be used for the longest time in future.

Step-by-Step Solution

Step	Page	Frame Contents	Hit/Fault
1	0	0	Fault
2	1	0 1	Fault
3	2	0 1 2	Fault
4	3	3 1 2	Fault
5	3	3 1 2	Hit
6	2	3 1 2	Hit
7	1	3 1 2	Hit
8	0	0 1 2	Fault
9	4	0 1 4	Fault
10	0	0 1 4	Hit
11	5	5 1 4	Fault
12	1	5 1 4	Hit
13	6	6 1 4	Fault
14	2	2 1 4	Fault
15	3	3 1 4	Fault

Total Page Faults (Misses)

Count faults:

```
[  
\boxed{10}  
]
```

Total Page Hits

Count hits:

```
[  
\boxed{5}  
]
```

Hit Ratio

Formula:

```
[  
\text{Hit Ratio} = \frac{\text{Hits}}{\text{Total References}}  
]  
[  
\frac{5}{15}  
]  
[  
0.333  
]  
[  
\boxed{33.3%}  
]
```

Miss Ratio

Formula:

```
[  
\text{Miss Ratio} = \frac{\text{Misses}}{\text{Total References}}  
]  
[  
\frac{10}{15}  
]  
[  
0.666  
]  
[  
\boxed{66.6%}  
]
```

Final Answer

Parameter	Value
Total Hits	5
Total Misses/Page Faults	10
Hit Ratio	33.3%

Parameter	Value
Miss Ratio	66.6%

Q.5 Find the miss and hit ratios using the Optimal Page Replacement Algorithm

Given

Frame Size

[
4
]

Reference String

8, 9, 0, 9, 8, 3, 0, 1, 0, 0, 1, 1, 8, 9, 0, 3

Optimal Page Replacement Algorithm

In the Optimal Page Replacement Algorithm:

- The page that will not be used for the longest time in future is replaced.
-

Step-by-Step Solution

Step Page Frame Contents Hit/Fault

1	8	8	Fault
2	9	8 9	Fault
3	0	8 9 0	Fault
4	9	8 9 0	Hit
5	8	8 9 0	Hit
6	3	8 9 0 3	Fault
7	0	8 9 0 3	Hit
8	1	8 1 0 3	Fault
9	0	8 1 0 3	Hit
10	0	8 1 0 3	Hit
11	1	8 1 0 3	Hit
12	1	8 1 0 3	Hit
13	8	8 1 0 3	Hit
14	9	9 1 0 3	Fault
15	0	9 1 0 3	Hit
16	3	9 1 0 3	Hit

Total Page Faults (Misses)

[
\boxed{6}
]

Total Page Hits

[
\boxed{10}
]

Hit Ratio

Formula:

```
[  
\text{Hit Ratio} = \frac{\text{Hits}}{\text{Total References}}  
]  
[  
\frac{10}{16}  
]  
[  
0.625  
]  
[  
\boxed{62.5%}  
]
```

Miss Ratio

Formula:

```
[  
\text{Miss Ratio} = \frac{\text{Misses}}{\text{Total References}}  
]  
[  
\frac{6}{16}  
]  
[  
0.375  
]  
[  
\boxed{37.5%}  
]
```

Final Answer

Parameter	Value
Total Hits	10
Total Misses/Page Faults	6
Hit Ratio	62.5%
Miss Ratio	37.5%